

Crowd Counting on Images with Scale Variation and Isolated Clusters

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Introduction

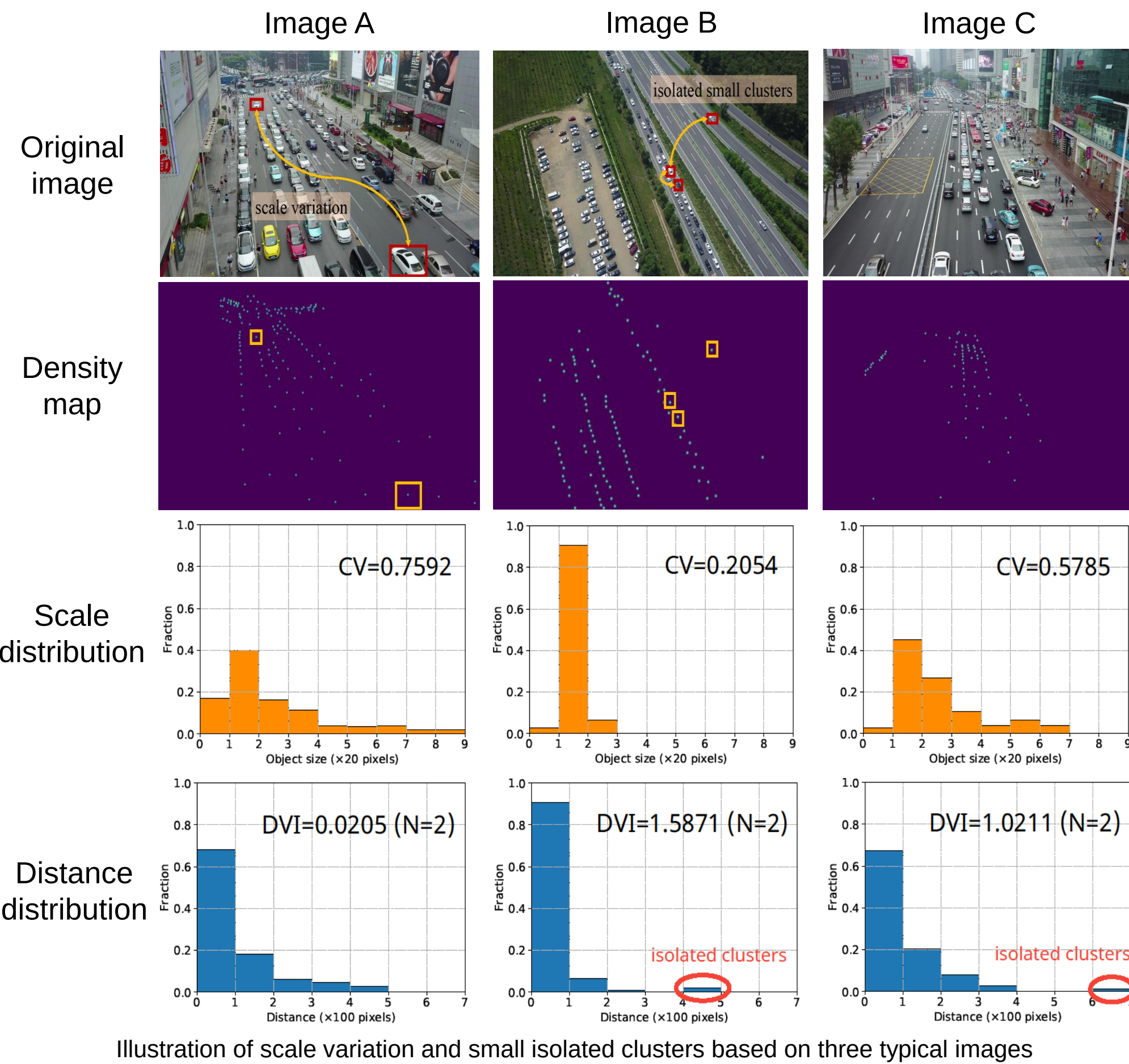
Wide applications:

- Public safety
- Traffic management
- Parade crowd analysis

Approach (SACANet):

- Scale-Adaptive long-range Context-Aware Network
- Tackle large scale variation problem
- Address Isolated object clusters problem

Challenges



Precise crowd counting remains challenging, mainly due to:

- Large variation in object scale.** Due to the perspective and distance of the camera with respect to an object, two objects of the same physical size would appear as different size in an image.
- Isolated small clusters of objects.** An image often has isolated object clusters. These isolated clusters may be many but small in size.

Typical Images Illustration

Measurement:

- Coefficient of variation (CV) to measure the level of scale variation. The larger the CV is, the more challenging the scale variation problem is.
- Dunn Validity Index (DVI) to measure the level of isolated clusters. The larger the DVI, the more small-size scattered clusters there are.

Illustration:

- Large scale variation without severe isolated clustering (Image A).
- Isolated clustering without severe scale variation (Image B).
- Both scale variation and isolated clustering (Image C).

Crowd Counting for Drone-based Scenes

Drone sensors and surveillance both capture crowd scenes. Drone-based crowd counting has not yet been fully explored, and it also lacks publicly large-scale diversified datasets.

Statistics of the newly proposed VisDrone2019 People & Vehicle crowd counting datasets

Dataset	Average Resolution	Images	Max	Min	Total
People	969×1482	3347	289	10	108,464
Vehicle	991×1511	5303	349	10	198,984

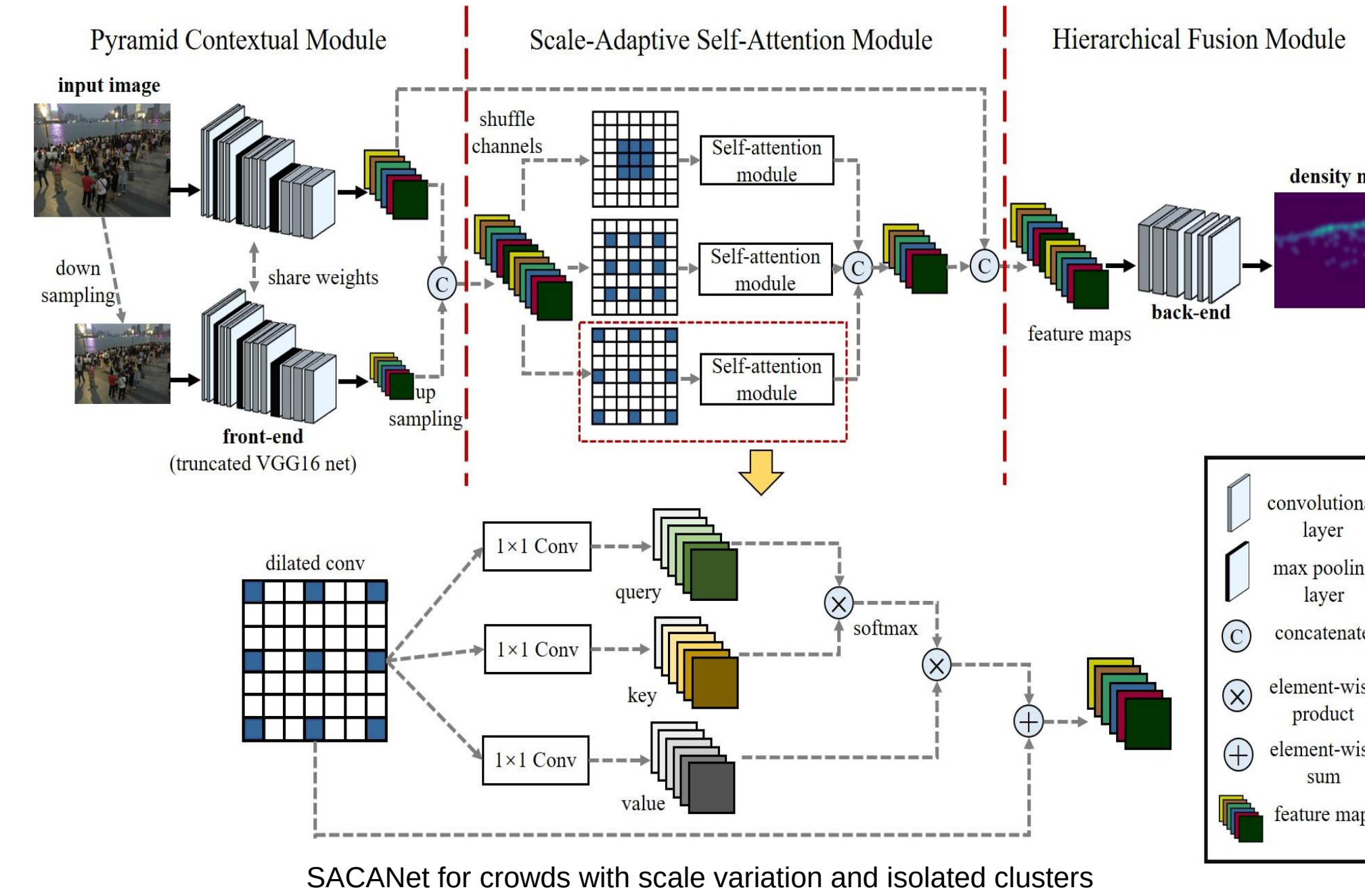
We modified the VisDrone2019 challenging dataset into two large-scale diversified drone-based crowd counting datasets, from the bounding boxes of targets to crowd counting annotations.

- VisDrone2019 people.** The new People annotation location is the head point of the original bounding box.
- VisDrone2019 vehicle.** The new vehicle annotation location is the center point of the original bounding box. The two kinds of operations are defined as follows:

$$People[cols, rows] = [bb_{left} + \frac{bb_{width}}{2}, bb_{top}],$$

$$Vehicle[cols, rows] = [bb_{left} + \frac{bb_{width}}{2}, bb_{top} + \frac{bb_{height}}{2}].$$

Methods



The pyramid contextual module.

This module aims to extract long-range contextual information and enlarge the receptive field.

The scale-adaptive self-attention module.

It is to accommodate the wide scales and detect isolated clusters via the long-range contextual information. The module first transfers input x to query Q_x , key K_x and value V_x : $Q_x = f(x)$, $K_x = g(x)$, $V_x = h(x)$. The output weighted density map Y is computed by two kinds of matrix multiplications:

$$Y = \text{softmax}(Q_x K_x^T) V_x.$$

The hierarchical fusion module.

The module integrates multi-level self-attention features to achieve accurate crowd counting estimation.

Group normalization for better convergence.

This has no relation to batch size and enables our network to converge.

Objective function.

We use the Euclidean loss to optimize the model:

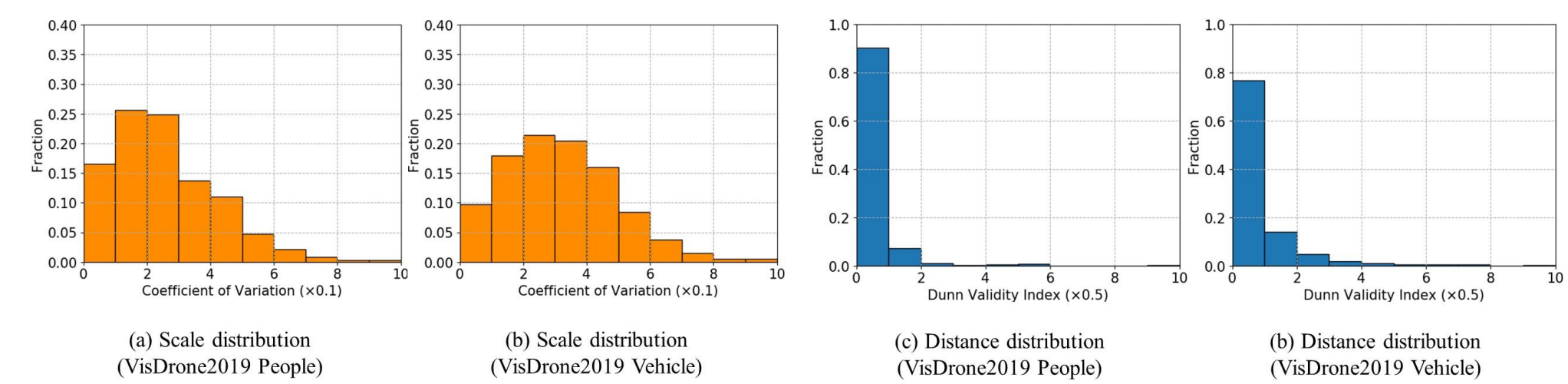
$$L_E = \frac{1}{N} \|F(X; \alpha) - Y\|_2^2.$$

Results

Results on the VisDrone2019 datasets

Quantitative results on the VisDrone2019 datasets					Ablation study on the VisDrone2019 datasets			
Method	People		Vehicle		Method	People		Vehicle
	MAE	MSE	MAE	MSE		MAE	MSE	
VGG-16	22.0	44.8	21.4	29.3	baseline	14.5	40.8	12.5
MCNN	16.4	39.1	14.9	21.6	baseline+context	13.8	39.2	11.0
CSRNet	12.1	36.7	10.9	16.6	baseline+context+SASA	11.7	36.1	9.1
SACANet (ours)	10.5	35.1	8.6	12.9	SACANet (ours)	10.5	35.1	8.6

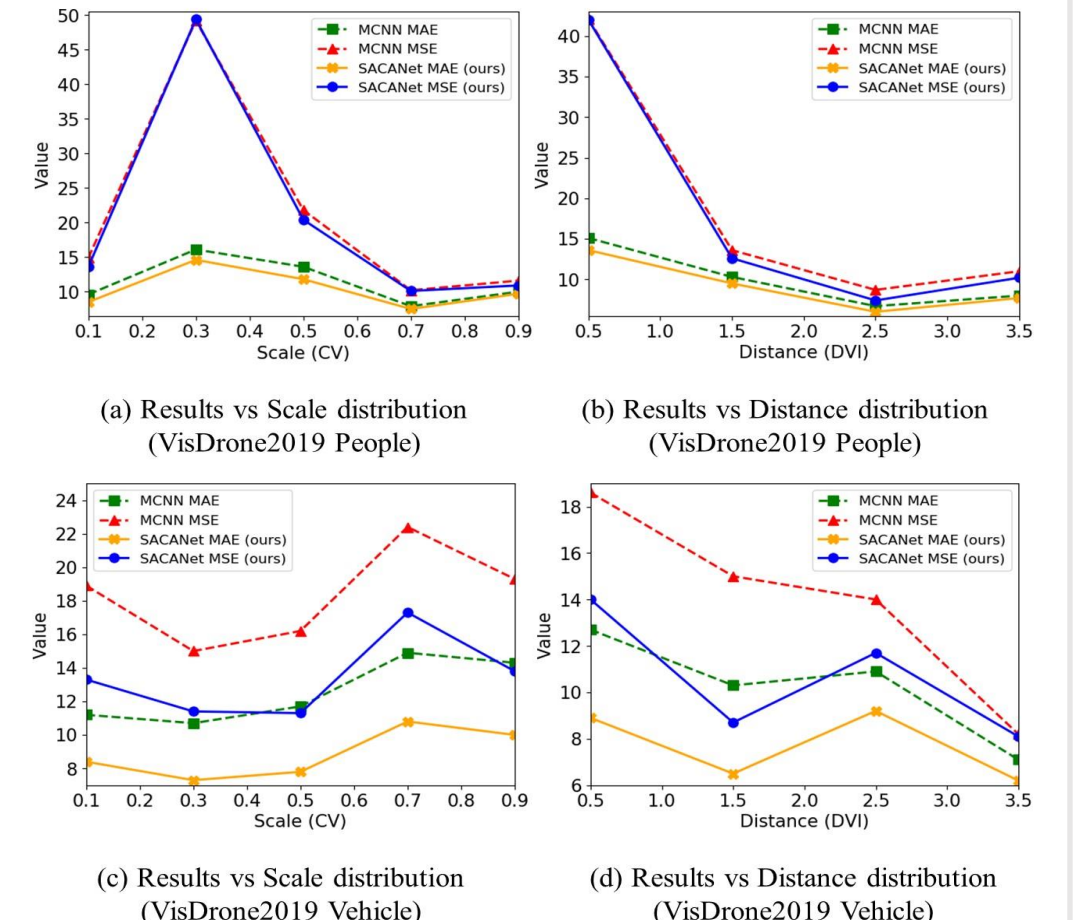
Distributions on VisDrone2019 Vehicle & People dataset



Evaluation on ShanghaiTech A & B

Results on the ShanghaiTech A & B datasets				
Method	Dataset A		Dataset B	
	MAE	MSE	MAE	MSE
MCNN	110.2	173.2	26.4	41.3
Switching-CNN	90.4	135.0	21.6	33.4
CP-CNN	73.6	106.4	20.1	30.1
ACSCP	75.7	102.7	17.2	27.4
IG-CNN	72.5	118.2	13.6	21.1
CSRNet	68.2	115.0	10.6	16.0
DRSAN	69.3	96.4	11.1	18.2
SANet	67.0	104.5	8.4	13.6
Baseline	68.3	107.5	11.9	18.7
SACANet (ours)	64.4	95.9	7.8	13.5

Comparison via different distributions



Qualitative results on VisDrone2019 Vehicle & People dataset

